

Total No. of Questions : 8]

[Total No. of Printed Pages : 2

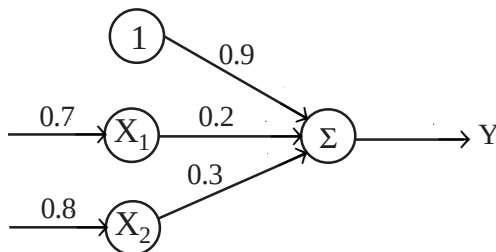
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**EE-7102 (GS)**  
**B.E. VII Semester Examination, June 2020**  
**Grading System (GS)**  
**Soft Computing Techniques and Applications**  
**Time : Three Hours**

**Maximum Marks : 70**

**Note:** i) Attempt any five questions.  
ii) All questions carry equal marks.

1. What is Monte-Carlo simulation technique? How does the Monte-Carlo simulation work? State the benefits of Monte-Carlo simulation technique.
2. a) Discuss the concept of binary and Real Parameter of Genetic Algorithms.  
b) Explain constraints handling in GA.
3. Draw the general architecture and write a training algorithm of Radial Basis Function network (RBF).
4. A random variable X has the density function
$$f(x) = \frac{C}{(x^2 + 1)} \text{ where, } -\infty < x < \infty$$
  - i) Find the value of the constant C.
  - ii) Find the probability that  $X^2$  lies between  $\frac{1}{3}$  and 1.
5. Explain back propagation in detail.
6. Calculate the output of the network shown in figure. Use binary and bipolar sigmoidal functions.



7. Explain the following W.R.T neural networks:
  - i) Input layer
  - ii) Hidden layer
  - iii) Out put layer
  - iv) Weights
  - v) Threshold

[2]

8. Write short notes on any two of the following:

- a) Genetic neuro hybrid system
- b) Particle swarm optimization and its variants
- c) Functional link network
- d) Optimization of economic dispatch problem using two-members non-recombinative ES.

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